European Integration and Urban Growth

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Abstract

Spatial inequality within countries is high and rising. This paper assesses how much the 2004 enlargement of the European Union contributed to widening disparities in economic performance across cities in Europe. I develop a quantitative open-economy model of economic geography in which market integration spurs city growth. Trade occurs through urban hubs and generates positive spillovers for their local consumer amenities. As a result, removing barriers to trade causes economic activity and population to concentrate closer to urban hubs. I show that empirical evidence bears out qualitatively the theoretical predictions of my model. Quantitatively, my structural estimates imply that greater market integration caused by EU enlargement accounts for 29.6% of the observed increase in residents of the major cities in Central Europe.

Keywords: European integration, border effects, economic geography.

1 Introduction

Spatial inequality within developed countries has been rising over the last few decades. This phenomenon is particularly well documented for the United States, as large cities continue to attract an ever growing number of skilled workers (Berry and Glaeser, 2005) (Moretti, 2012) and diverge from their smaller and more peripheral counterparts in terms of economic performance (Giannone, 2017). Western Europe has experienced similar trends, with rising disparities among metropolitan areas since about 2004 (Ehrlich and Overman, 2020). The diverging fortunes of Europe's cities and concomitant political, social and economic consequences are becoming a growing cause for concern (Rodríguez-Pose, 2018).

Concurrent with this rise in spatial disparities; Europe has also undergone a dramatic process of market integration, that culminated with the 2004 enlargement of the European Union as its keystone. Thereafter, eight Eastern European countries along with Cyprus and Malta joined the union and became permanent members of the single market.[1] These two major developments in European economic geography may well be connected. Theoretically, the link between market access and agglomeration is a seminal result in economic geography (Krugman, 1991). Empirically, however, the question remains open: How much did European integration contribute to the diverging fortunes of European cities?

This paper investigates to what extent urban divergence in Europe and the growing pre-eminence of regional superstar cities can be explained by market integration. I build a quantitative model of economic geography that connects trade and city growth. The model delivers predictions that are consistent with the data, where I observe differentially higher growth in cities that gain more in terms of market access. I structurally estimate my model to measure the effects of the 2004 enlargement of the European Union, where I treat the accession of new member states into the union as a reduction in border related trade barriers between the old (EU15) and new member states (NMS). I find that the EU enlargement can explain 29.6% of the rise in population of the major cities in Central Europe.

My model economy features a large set of locations that differ in their amenities and their proximity to one another. Trade takes place only at a subset of locations that serve as trading hubs and generates positive spillovers in the form of local consumer amenities. Labor

[1] New member states had individual free trade agreements with the EU prior to their accession. However their admission into the EU still had a significant effect on trade costs via reductions in tariffs and non-tariff barriers. Furthermore, abolishing customs checkpoints between countries was a major milestone in increasing market integration and led to efficiency gains in terms of travel time.

is mobile across locations, subject to mobility frictions. The spatial equilibrium of my model entails a rich geography both across and within market areas that form endogenously around each hub. As trade opportunities increase, the cost of distance from the hub becomes more salient. As a result, workers move closer to the hub through which they trade. Residents of the hub cities also benefit from consumer amenities that grow larger as trade expands. This endogenous increase in consumer amenities amplifies the rise in *regional urban primacy*, or the concentration of population that reside in the trading hub within each market area.

Besides this overall increase in urban primacy, the model predicts heterogeneous impacts across hubs, depending on the increase in their market access. Hubs located close to the border between the NMS and the EU15 such as Berlin, Vienna and Ljubljana reap the largest gains in market access and thrive. In contrast, Southern Italian hubs such as Naples and Palermo become even more peripheral and suffer from trade diversion. In general equilibrium, they may even lose population as economic activity reallocates across market areas towards those that benefit the most from the eastward expansion of the union.

Identifying the relevant urban hubs is the starting point for my quantitative empirical analysis. I rely on the algorithm developed by Mori, Smith, and Hsu (2020) to construct a hierarchical city system for each country by recursively partitioning its geography based on the location and the size of its largest cities. This procedure selects 73 hubs from the total sample of 304 cities across 8 countries.[2] Having selected the hubs, I establish two patterns in the data that are consistent with the predictions of my model. First, annual population growth among hubs increased by roughly 0.4 percentage points relative to non-hub locations outside the designated border zone. Second, hubs within 50 kilometers of the accession border show a suggestive additional increase of 0.5 percentage points in annualized population growth, significant at the 10% level.[3]

In light of these empirical facts, I structurally estimate the model to quantify the effects of the 2004 Enlargement of the EU on the spatial distribution of economic activity and population. I calibrate my model to match the hub and country population levels before EU enlargement. Although I match only hub and country population totals, the correlation between model-implied and data-based 2001 populations is 0.952 across the 36,669

[2] I use the set of cities defined by Eurostat as candidates for hubs; their geographic representation is described in Section 3.2.

[3] In my quantitative and empirical analysis, I specifically focus on the set of countries that are contiguous to the border between the (EU15) and the (NMS). Accession of the NMS into the union did have transformative effects on the continent as a whole. Even so, shifts in relative market access generated by the integration of Eastern European economies into the single market were strongest within the geography I consider.

municipalities in 8 countries that make up my full sample; it is 0.488 among the 36,328 municipalities outside the selected hubs.[4] To isolate the effects of market integration, I then solve for the counterfactual scenario in which all structural parameters of the model are kept fixed except border-related barriers to trade.

My results show how economic integration contributed to the reshaping of European economic geography. In my counterfactual, the aggregate population of all urban hubs grows by 0.465%, equivalent to 29.6% of the 1.57% increase observed in the data. However, my counterfactual abstracts from changes in overall population. The share of total population living in all hubs increases by 0.1004 percentage points, exceeding the observed increase of 0.0768 percentage points. Turning to the effects on real income, I find that all countries gain at the aggregate level, and the population-weighted average local increase in real income is 0.22%. Overall, new member states are the biggest winners from the enlargement process. On average, their real income increases by 0.4%. Since Eastern European countries are initially poorer, enlargement entails a decrease in overall income inequality. The population-weighted standard deviation of real income across locations in my sample falls by 0.4%.

My model also speaks to the evolution of within-country spatial inequality. In Germany, the East–West income gap narrows, as real income growth in East Germany (0.17%) is considerably higher than in the West (0.1%). Integration of Eastern European markets improves the centrality of East German hubs within Europe's trade network relative to the West, allowing them to catch up. Conversely, European integration exacerbates the long-standing economic dichotomy in Italy between the North and South. Southern Italian hubs become even more peripheral relative to their Northern counterparts within the new single market. As a result, the North continues to pull away.

My work is related to the extensive literature that studies the effects of trade on local economic outcomes such as population (Redding and Sturm, 2008; Bleakley and Lin, 2012), employment and wages (Brülhart, Carrere, and Trionfetti, 2012; Brülhart, Carrère, and Robert-Nicoud, 2018; Caliendo, Dvorkin, and Parro, 2019), regional development and inequality (Dix-Carneiro and Kovak, 2017; Brooks, Gendron-Carrier, and Rua, 2018) and country development (Pascali, 2017; Feyrer, 2019). Port development illustrates that local costs can offset the population gains from improved trade access (Ducruet et al., 2024). It is also related to the literature on the determinants of urban growth and divergence such as

[4]Population data come from Eurostat historical population series, harmonized to municipality polygons; missing values used for model accounting are treated as described in Section 3.3.

skill composition (Berry and Glaeser, 2005; Shapiro, 2006; Moretti, 2012; Diamond, 2016; Ganong and Shoag, 2017), institutional quality (Henderson and Wang, 2007), technological change (Henderson and Wang, 2005) and trade openness (Krugman and Elizondo, 1996). Recent work also emphasizes the roles of human capital and urban costs in city growth (Duranton and Puga, 2023). Bakker et al. (2024) show that export activity is disproportionately concentrated in larger cities and quantify how trade liberalization reallocates employment toward them. I study how the 2004 EU enlargement changed population concentration around trading hubs and assess the additional role of trade-driven amenities.

My paper fits in a broader literature on quantitative spatial economics. Redding and Rossi-Hansberg (2017) provide an extensive review of this rapidly growing literature. One strand of this literature proposes models in which trade contributes to urban growth via different mechanisms (Coşar and Fajgelbaum, 2016; Fajgelbaum and Redding, 2022; Bakker et al., 2024). Nagy (2023) further shows how trading routes and agricultural hinterlands shape city formation and growth. My paper contributes to this literature by developing a quantitative spatial economic framework based on Nagy (2022), who studies the effects of the redrawing of Hungary's borders after World War I on its economic geography. My model extends this framework to incorporate multiple countries. I do not assume, as Nagy (2022) did, that borders are prohibitive to trade. Instead, I estimate the magnitude of border-related barriers before and after 2004 and measure the effects of this change on the spatial distribution of economic activity.

Moreover, my model incorporates trade-driven endogenous amenities in urban hubs, which Nagy (2022) did not consider. A growing body of literature highlights the importance of local consumption amenities in spurring urban growth (Glaeser, Kolko, and Saiz, 2001; Rappaport, 2007; Carlino and Saiz, 2008; Diamond, 2016). Leonardi and Moretti (2023) show that restaurants agglomerate and local amenities diverge across neighborhoods in Milan. Prior work mainly related endogenous amenities to a city's population or skill composition. In this paper, I focus on a different determinant: economic activity arising from trade. Intuitively, trade makes a city more cosmopolitan and facilitates its development as a center of culture, art and science. Furthermore, well-connected hubs attract a greater variety of brands and a more diverse mix of people, both of which are not available in smaller, more peripheral locations. Thus, urban hubs provide exclusive access to a wide range of activities and goods, and also contain a richer social fabric. Trade enhances their capacity to provide such services and generates spillovers in the form of consumer amenities.

My analysis also complements studies that focus on the economic implications of the EU enlargement (Baldwin et al., 1997; Henrekson et al., 1997; Dustmann and Frattini, 2012; Kennan, 2017; Caliendo et al., 2021). I further this literature by assessing the impact of European integration on differential city growth.

The rest of the paper is organized as follows. Section 2 develops the quantitative model. Section 3 describes its calibration and structural estimation. Section 4 discusses the model's predictions and presents supportive evidence. Section 5 presents the results of the quantitative exercise. Section 6 concludes.

2 A Model of Trade and Urban Growth

The model links lower trade barriers to population concentration around trading hubs. Workers produce at their residential locations and trade through a hub, paying a distance-related utility cost to reach it. Better market access raises real income and strengthens the incentive to live near the hub. Trade also improves consumer amenities within hub cities, amplifying their population gains. Heterogeneous location preferences counterbalance these agglomeration forces.

The geography G consists of a finite set of locations $r \in G$ across countries C. Each worker has one unit of labor and produces a uniquely differentiated good using labor alone. Goods trade within and across countries, with additional costs at national borders. Trade takes place only at hub locations $\mu_i \in G$. Workers choose both a residential location r and a domestic trading hub h at which to sell their output and buy goods.

2.1 Consumption

Worker i resides at location r, trades through hub h, and obtains utility

$$u_h(r,i) = a(r,i) + \zeta(h,r)^{-1} \left[\sum_{j=1}^{\bar{L}} c_h^j(r,i)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \tag{1}$$

Here, $a(r,i)$ denotes residential amenities, and $\zeta(h,r) \geq 1$ is the utility cost of shipping between the residence and the trading hub. Its inverse discounts consumption utility. The quantity $c_h^j(r,i)$ is worker i's consumption of worker j's good, $\bar{L}$ is total population, and

$\sigma > 1$ is the elasticity of substitution across goods.

Residential amenities combine a component common to all workers with an idiosyncratic location preference:

$$a(r,i) = a(r) + \epsilon(r,i) \tag{2}$$

where $a(r)$ is the common component and $\epsilon(r,i)$ captures worker-specific tastes. I assume that $\epsilon(r,i)$ is independent across workers and locations and follows a Gumbel distribution:

$$F(\epsilon(r,i) \leq z) = e^{-e^{-\frac{z}{\theta}}} \tag{3}$$

where θ is the scale parameter governing taste heterogeneity. Larger θ weakens the response of location choices to common amenities and consumption incentives. As $\theta \to \infty$, these incentives become negligible relative to idiosyncratic tastes, and workers distribute equally across the model's discrete locations.

The common amenity component has exogenous and trade-related parts:

$$a(r) = \bar{a}(r) + \tilde{a}(r) \tag{4}$$

where $\bar{a}(r)$ represents exogenous amenities, such as climate, and $\tilde{a}(r)$ represents amenities generated by trade.

Trade-related amenities satisfy

$$\tilde{a}(r) = T(r)^{\alpha} \tag{5}$$

where $T(r)$ is trade conducted at location r and α is the elasticity of trade-related amenities with respect to trade. Section 3.5 describes the amenity calibration and the identifying moment for α.

The residence-to-hub shipping cost increases exponentially with distance:

$$\zeta(\mu(r), r) = e^{\psi \cdot dist_{\mu(r),r}}$$

The parameter ψ governs the increase in this cost per kilometer. Workers prefer to live near their chosen hub to reduce shipping costs. I assume that residence-to-hub shipping across

national borders is prohibitive, so each worker chooses a hub in her country of residence. Goods can still trade internationally between hubs.

2.2 Production

Each worker uses her unit of labor to produce one unit of output at her residence and sells it through her chosen hub. Under monopolistic competition, worker i chooses her product price p_h^i, taking the hub price index P_h and inter-hub transport costs $\tau(\mu_h, \mu_o) \geq 1$ as given. I assume symmetric transport costs, $\tau(\mu_h, \mu_o) = \tau(\mu_o, \mu_h)$, with

$$\tau(\mu_h, \mu_o) = e^{\phi \cdot dist_{\mu_h, \mu_o} + \mathbb{1}_{h,o}\beta_{h,o}}$$

where ϕ is the change in log transport costs per kilometer, $\mathbb{1}_{h,o}$ indicates hubs in different countries, and $\beta_{h,o}$ is the log border-cost term. Exponential transport costs are common in trade and economic geography models (Head and Mayer, 2014; Desmet and Rossi-Hansberg, 2014). The border term captures tariff and non-tariff impediments to trade (Santamaría et al., 2021), including legal differences and uncertainty (Turrini and van Ypersele, 2010) and frictions in business and social networks (Combes et al., 2005).

Inter-hub transport costs favor hubs with better market access, while residence-to-hub shipping costs favor living close to those hubs. Idiosyncratic location preferences limit population concentration.

2.3 Equilibrium

Amenities depend on the residential location rather than the chosen trading hub, and utility is additively separable in amenities and consumption. All workers at a given location therefore choose the same hub, denoted by $\mu(r)$. A spatial equilibrium satisfies the following conditions.

Definition 1 *Given a geography G, parameters $\sigma, \theta, \phi, \psi, \beta, \alpha, \bar{L}$ and functions $a : G \to \mathbb{R}_+$, $\{\tau, \zeta\} : S^2 \to \mathbb{R}_+$ an equilibrium of this economy consists of a population distribution $L : G \to \mathbb{R}_+$, consumption levels $c : [0, \bar{L}] \times G \times [1, ..., M] \to \mathbb{R}_+$, goods prices and levels of production $p, x : [0, \bar{L}] \times [1, ..., M] \to \mathbb{R}_+$; and a function that assigns each residential location a trading hub $\mu : G \to [1, ..., H]$, such that following conditions hold:*

1. *Workers choose their consumption, production, price, residential location and trading hub to maximize their utility subject to the production technology and their budget constraint.*

2. *Goods markets clear at every hub, implying*

$$x_h^i = \sum_o \tau_{h,o}^{1-\sigma} (p_h^i)^{-\sigma} P_o^{\sigma-1} p_o L_o \tag{6}$$

for each worker i, where x_h^i is output sold through hub h, and P_o, p_o and L_o are the price index, common product price and number of workers trading through destination hub o.

In equilibrium, the number of people residing in location r is given by

$$\log L(r) = v + \theta^{-1}\left[a(r) + \zeta(\mu(r), r)^{-1} MA_{\mu(r)}^{\frac{2\sigma-1}{\sigma(\sigma-1)}}\right] \tag{7}$$

where $L(r)$ is residential population, v is the common location-choice normalizer, and $MA_{\mu(r)}$ is the market access of the chosen hub. Higher amenities and better hub market access raise a location's attractiveness. Holding amenities and hub market access fixed, rising shipping costs generate a negative population gradient away from the hub.[5]

Given their location choice, workers prefer the hub that offers them the best combination of proximity and market access.

$$\zeta(\mu(r), r)^{-1} MA_{\mu(r)}^{\frac{2\sigma-1}{\sigma(\sigma-1)}} \geq \zeta(\mu_m, r)^{-1} MA_m^{\frac{2\sigma-1}{\sigma(\sigma-1)}} \quad \forall m \tag{8}$$

Market access of any trading hub h is implicitly defined by

$$MA_h = \sum_o MA_o^{-\frac{\sigma-1}{\sigma}} \tau_{h,o}^{1-\sigma} L_o \tag{9}$$

Hub h has better market access when transport costs $\tau_{h,o}$ to large trading hubs are low. Appendix B shows that hub real income, $\omega_h = p_h/P_h$, increases with market access:

$$\omega_h = MA_h^{\frac{2\sigma-1}{\sigma(\sigma-1)}} \tag{10}$$

[5]The partial derivative of equation 7 with respect to distance, holding amenities, market access and the normalizer fixed, is $\frac{\partial \log L(r)}{\partial dist_{\mu(r),r}} = -\psi\theta^{-1}\zeta(\mu(r), r)^{-1} MA_{\mu(r)}^{\frac{2\sigma-1}{\sigma(\sigma-1)}} < 0$ for $\psi > 0$.

The number of workers trading through hub h is

$$L_h = \sum_{r:h=\mu(r)} L(r) \tag{11}$$

Equations (7)–(11) determine hub market access, hub choice and residential population jointly. Workers prefer locations near hubs with good market access, but heterogeneous location preferences prevent all population from concentrating at a single hub.

Following Nagy (2022), I define a local *urban-primacy index* as the response of log population to residence-to-hub proximity, measured by inverse shipping costs. The index holds amenities, hub assignments and the location-choice normalizer fixed:

$$UI(r) = \frac{\partial \log L(r)}{\partial \left[\zeta(\mu(r), r)^{-1}\right]}$$

A larger index means a steeper population gradient with respect to inverse shipping costs. Equation 7 gives

$$UI(r) = \theta^{-1} M A_{\mu(r)}^{\frac{2\sigma-1}{\sigma(\sigma-1)}}$$

The index increases with hub market access. Higher real income makes the consumption cost of distance more consequential, strengthening the incentive to live near the hub. This conditional gradient describes the model's agglomeration mechanism; the counterfactual population share in hub cores also depends on amenity changes and general equilibrium reallocation.

3 Structural Estimation

3.1 Data

The analysis combines Eurostat historical municipality populations, European trade flows and German city-level employment and housing data. Population data determine hub selection, calibrate the spatial model and measure population growth. Trade data identify the distance and border-cost parameters. German employment and housing data provide supporting evidence on the trade-amenity channel.

Eurostat's historical population series covers the start of each decade from 1961 to 2011. The main sample consists of 36,669 municipalities or historical local units in Austria, the Czech Republic, Germany, Hungary, Italy, Poland, Slovakia and Slovenia, the eight countries along the EU15–NMS accession border.[6] I use 2001 population to calibrate the baseline model and 1991, 2001 and 2011 population to construct the two growth periods in the reduced form analysis.

For post-enlargement border effects, I use regional freight flows constructed by Santamaría et al. (2021) from Eurostat's European Road Freight Transportation (ERFT) Survey. Santamaría et al. (2023) use these flows to document distance effects and home and country biases in European trade. Comparable regional flows are unavailable before enlargement, so the pre-enlargement estimates use country-level COMTRADE flows and CEPII bilateral controls. Section 3.6 explains how this difference in geographic coverage affects the comparison.

For the German amenity analysis, I combine sectoral employment from the German Regional Statistics Office with Eurostat city-level housing data.[7] Employment is measured at the NUTS3 district level across two-digit WZ93 industries. I construct a correspondence between districts and Eurostat cities; city and district boundaries coincide for most matched cities. The final sample contains 64 cities, including 19 selected hubs, with usable rent, wage and export-exposure data.

3.2 Determining Hubs

The model imposes trade through a subset of cities rather than estimating each city's intermediary trade flows. This assumption is motivated by evidence that exports concentrate in larger cities (Bakker et al., 2024), central cities ship disproportionately to their hinterlands (Mori and Wrona, 2024), and hinterlands shape city formation (Nagy, 2023). I select hubs using population and spatial hierarchy, then test whether the selected cities display the growth patterns predicted by the model.

I first construct the candidate city set using Eurostat's degree-of-urbanization categories: cities, towns and suburbs, and rural areas. In Germany, Slovenia, Hungary and the Czech Republic, the candidates are municipalities classified as cities. In Italy, Poland and Slovakia,

[6] Slovenia contributes 58 historical LAU1 units.

[7] Employment dataset: 42111-01-02-4. Housing dataset: urb_clivcon.

I also aggregate municipalities into greater cities where those boundaries are available: Milan, Naples, Katowice, Bratislava and Kosice.[8] Greater-city boundaries capture urban cores that extend across municipal borders.

I select hubs with the hierarchical partition algorithm of Mori et al. (2020). At each layer, the algorithm partitions each region around its K largest cities. I set $K = 3$ and stop the country-wide recursion before any parent region would contain fewer than three candidate cities for the next layer.

Layer 0 is the entire country, centered on its largest city. Layer 1 is a Voronoi-3 partition around the three largest cities; layer 2 partitions each of those regions around its three largest cities, and so on.[9] A complete layer k has 3^k centers. I retain terminal centers with populations above 150,000 as trading hubs. The common population threshold makes hub sizes more comparable across countries.

Figure 1 illustrates Germany's first two layers. Panel A partitions the country around Berlin, Hamburg and Munich. Panel B partitions each of those regions into three, yielding nine centers.

[8] I map Eurostat's 2014 Urban Audit city and greater-city boundaries onto the historical municipality geography. A municipality is assigned when at least 25% of its area overlaps a city footprint, with greater cities taking priority. Contiguous members are merged using a 100-meter tolerance; disconnected components remain separate candidates. The 304 city or greater-city identifiers produce 305 spatial components because Naples has a disconnected component.

[9] A Voronoi region contains the points closer to its center than to any other center. Assignments stay within the parent region and use distances between representative points on the candidate polygons.

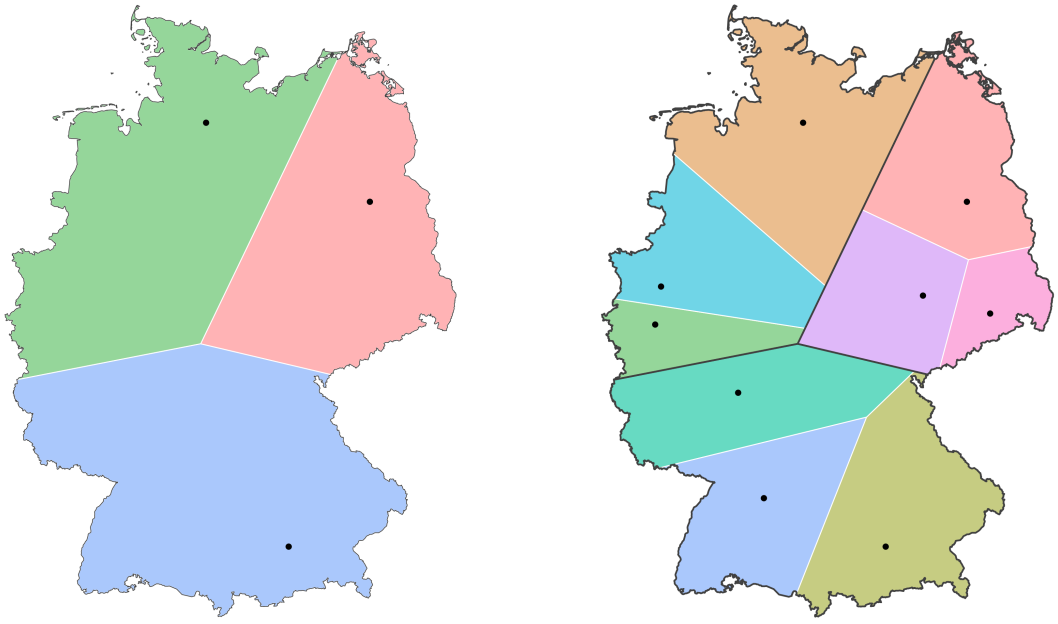


Figure 1: Spatial hierarchical 3-partition of Germany: (A) Layer-1 with 3 hubs (B) Layer-2 with 9 hubs.

Figure 2 shows the 73 selected hubs; Appendix A.2 lists them. Slovenia stops at layer 0; Austria, Hungary and Slovakia at layer 1; and the Czech Republic at layer 2. Germany, Poland and Italy reach layer 3. Italy's terminal implementation contains 26 rather than 27 centers before applying the population threshold. The final 73 hubs are selected from 304 candidate city identifiers.[10]

3.3 Model Geography

I use a fine discretization of space when setting up the model geography. I intersect the actual municipality polygons with a common-origin 1 km by 1 km square grid in the equal-area EPSG:3035 projection. Each positive-area municipality–grid intersection is a separate model location, retaining its municipality identifier, hub membership and land area. The 36,669 municipalities generate 1,924,743 locations. Hub footprints are the unions of their member municipality polygons. I use a point on the surface of each intersection to approximate its

[10]The population threshold excludes some terminal centers in Germany, Italy, Poland and the Czech Republic.



Figure 2: Urban Hubs

distance to each domestic hub and assign one market choice to the whole intersection.[11] Residential population is weighted by intersection area: for a piece of area A_r, the numerical version of equation 7 includes $\log A_r$ on the right-hand side. Thus the numerical logit allocates population per unit of land rather than giving equal weight to pieces of different sizes.

For location-to-hub distances, I use geodesic distances between representative points, with commuting restricted to domestic hubs. For hub-to-hub trade distances, I use OpenRouteService road-network distances for heavy goods vehicles, allowing ferry crossings. I check island connections and unusually long water-related detours against shortest truck-and-ferry routes and otherwise retain the default truck routes. I compute both directions and use their arithmetic mean.[12]

Hub and country population targets are aggregated from the retained historical municipality series. Missing model population values are filled using the geometric mean of positive 1991 and 2011 values for 2001 where available, then country-median decade growth, and finally the nearest available positive historical population; any remaining missing or non-positive value receives one person. This affects 74 municipalities in 2001 and 75 in 2011. These imputations are confined to model accounting and are not used to fill missing regression outcomes.[13]

3.4 Calibration of the Structural Parameters

I set the elasticity of substitution to $\sigma = 5$, following Simonovska and Waugh (2014), and the residence-to-hub shipping-cost parameter to $\psi = 0.1$, following Nagy (2022).

The parameter that governs heterogeneity in location preferences, θ, plays a central role in the effects of trade on urban primacy. Let r_h denote the centre of hub h, where $\zeta(h, r_h) = 1$. Holding amenities, hub assignments and the location-choice normalizer fixed, the direct income elasticity of population at that location is

[11] Numerical slivers no larger than 10^{-8} km^2 are dropped; every municipality retains at least one location. This discretization does not preserve within-piece variation in commuting costs, and no resolution-convergence claim is made.

[12] The routes use the September 2026 OpenStreetMap network accessed through OpenRouteService (`https://openrouteservice.org/`). I construct the full European hub matrix, including the United Kingdom, Switzerland and Norway, and extract the eight-country submatrix.

[13] The published analytical geography repairs the previously identified substantial municipality overlaps while retaining every population-backed ID. Reference areas without a population node remain outside the model and are tracked for cartography.

$$\left.\frac{\partial \log L(r_h)}{\partial \log \omega_h}\right|_{\omega_h=\bar{\omega}_h} = \left.\frac{\partial \log L(r_h)}{\partial \omega_h}\right|_{\omega_h=\bar{\omega}_h} \left.\frac{\partial \omega_h}{\partial \log \omega_h}\right|_{\omega_h=\bar{\omega}_h} = \theta^{-1}\bar{\omega}_h$$

Thus the direct elasticity increases with hub real income. This is a conditional response at the hub centre, not the general-equilibrium elasticity of total hub population. I choose θ using the unweighted mean of $\bar{\omega}_h/\theta$ across the selected hub centres. Estimates of income responsiveness in other settings include late-20th century Brazil (1.91) (Morten and Oliveira, 2018), present-day China (2.54) (Tombe and Zhu, 2019) and the present-day United States (3.3) (Monte, Redding, and Rossi-Hansberg, 2018). The calibration $\theta = 28.57$ yields an unweighted mean direct hub-centre elasticity of 2.50 in the reconstructed 2001 equilibrium.

I estimate the distance semi-elasticity of inter-hub transport costs, ϕ, using ERFT shipments and the model's gravity equation. Bilateral trade from hub h to hub o is

$$T_{ho} = \tau_{ho}^{1-\sigma} p_h^{1-\sigma} L_h P_o^{\sigma-1} p_o L_o \tag{12}$$

where T_{ho} is bilateral trade. Substituting the exponential transport-cost function and absorbing origin and destination terms into fixed effects gives

$$\log T_{ho} = \beta_1 dist_{ho} + \beta_2 B_{ho} + \delta_h + \delta_o + \epsilon_{ho} \tag{13}$$

Here, $dist_{ho}$ is distance in kilometers, B_{ho} indicates a national border, and δ_h and δ_o are origin and destination fixed effects. The distance coefficient satisfies $\beta_1 = (1-\sigma)\phi$. The reestimated raw-distance regression gives $\phi = 3.516 \cdot 10^{-4}$; the reported counterfactuals use the rounded calibration $\phi = 3.51 \cdot 10^{-4}$.[14]

3.5 Trade Related Amenities

Consumer amenities help explain city growth and residential location choices (Glaeser et al., 2001; Rappaport, 2007, 2008). Restaurants, cultural activities and other services can become concentrated within cities (Leonardi and Moretti, 2023). I extend the usual link between amenities and city size by allowing hub amenities to respond to trade. The calibration below identifies this response through a maintained orthogonality condition; the German city regressions provide separate, suggestive evidence.

[14] I retain $\sigma = 5$. The log-distance regression for border effects in Section 3.6 is a separate specification.

To calibrate the amenity elasticity α, I divide exogenous amenities into a country component $\bar{a}_C$ and an additional component $\bar{a}_h$ available only within hub h:

$$\bar{a}(r) = \begin{cases} \bar{a}_h + \bar{a}_C & \text{if } r \in h \\ \bar{a}_C & \text{otherwise.} \end{cases}$$

Because trade takes place through hubs, trade-related amenities are available only within hubs. At hub h, this component is $\tilde{a}_h = T_h^\alpha$, where T_h is total non-self trade conducted through the hub. Thus the total location-specific amenity is

$$a(r) = \begin{cases} \bar{a}_C + \bar{a}_h + \tilde{a}_h & \text{if } r \in h \\ \bar{a}_C & \text{otherwise.} \end{cases} \tag{14}$$

Write $a_h = \bar{a}_h + \tilde{a}_h$ for the additional hub amenity, above $\bar{a}_C$.

I first solve auxiliary equilibria for 2001 and 2011, allowing total hub amenities a_h and country amenities $\bar{a}_C$ to vary so that each equilibrium matches the observed hub and country population totals. Across hubs, changes in total amenities correlate negatively with changes in total non-self trade (-0.402), but positively with changes in trade across the EU15–NMS border (0.226). The endogenous amenity component depends on total non-self trade, T_h^α, whereas the identifying moment uses only opposite-bloc trade. For each candidate α, I recover residual exogenous amenities as $\bar{a}_h = a_h - T_h^\alpha$ and impose the moment condition

$$corr[\Delta \bar{a}_h, \Delta Trade_h^{E,W}] = 0 \tag{15}$$

where $\Delta \bar{a}_h$ is the change in residual exogenous hub amenities and $\Delta Trade_h^{E,W}$ is the 2001–2011 change in trade with hubs in the opposite bloc.[15] The identifying assumption is that changes in opposite-bloc trade are uncorrelated with changes in amenities outside the modeled trade channel. Accession being determined above the hub level does not by itself establish this condition: trade changes come from separately calibrated equilibria and also reflect observed population changes. Other local changes correlated with trade would violate the moment condition.

Figure 3 plots the moment as α varies. At $\alpha = 0$, amenities do not respond to trade,

[15] For western hubs in Austria, Germany and Italy, opposite-bloc trade is trade with the Czech Republic, Hungary, Poland, Slovenia and Slovakia, and conversely for eastern hubs.

so changes in residual exogenous amenities equal changes in total hub amenities and their correlation with opposite-bloc trade is 0.226. The correlation reaches zero near $\alpha = 0.195$, the value used in the eight-country counterfactual.

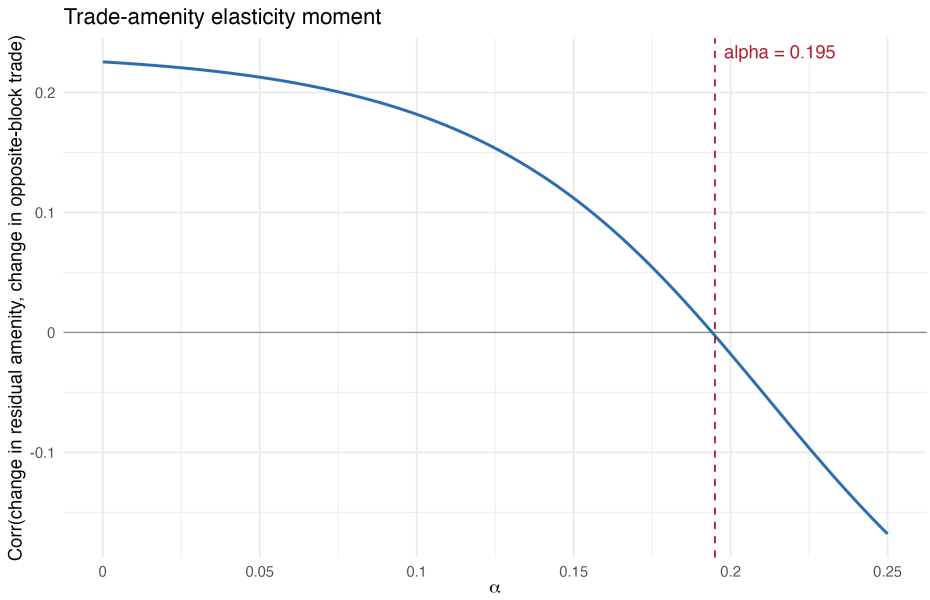


Figure 3: Estimation: Elasticity of Trade Driven Amenities

To provide supporting evidence for trade-related amenities, I examine whether cities with greater export exposure experienced higher rent growth relative to wage growth after a positive trade shock. If trade improves local amenities, housing demand and rents may rise relative to wages. Market access can also raise land prices relative to wages through production and nontraded-goods channels (Fajgelbaum and Redding, 2022); the rent–wage pattern is therefore suggestive rather than a direct identification of amenities. I measure rent growth $\Delta Rent_c$ and wage growth $\Delta Wage_c$ for German cities and define the outcome as $Y_{ct} = \Delta Rent_c - \Delta Wage_c$. Since city-level trade data are unavailable for Europe, I follow Dauth, Findeisen, and Suedekum (2014) and construct the export impact measure ExI_c from each city's initial sectoral employment composition. This measure captures exposure to increased demand for German exports from Eastern Europe.

$$ExI_{ct} = \sum_j \frac{L_{cjt}}{L_{jt}} \frac{\Delta EX_{jt}^{EAST}}{L_{ct}}$$

where ΔEX_{jt}^{EAST} is the change in German exports to Eastern Europe in industry j between t and $t+1$, in constant 2004 euros. The share L_{cjt}/L_{jt} is city c's share of German

employment in industry j, and L_{ct} is its total manufacturing employment. I fix the employment weights at their 2001 values to reduce feedback from subsequent trade growth into local industry composition:

$$ExI_{ct} = \sum_j \frac{L_{cjt-1}}{L_{jt-1}} \frac{\Delta EX_{jt}^{EAST}}{L_{ct-1}} \tag{16}$$

I relate this export-exposure measure to rent growth relative to wage growth, controlling for pre-period city characteristics:

$$Y_{ct} = \beta_1 ExI_{ct} + \beta' X_c + \epsilon_{ct} \tag{17}$$

The outcome Y_{ct} is rent growth minus wage-per-capita growth from 2005 to 2014. Export exposure covers 2004–2014, and X_c contains pre-period city characteristics. The baseline in Table 1 controls for 2001 dwellings per capita. Column 1 includes all 64 German cities with usable data; the exposure coefficient is small and imprecisely estimated. Column 2 includes 19 selected hubs. Among hubs, a €1,000 increase in export exposure per manufacturing worker is associated with 0.597 percentage points higher relative rent growth (SE 0.272; $p = 0.043$).[16] The pattern is consistent with the amenity mechanism, but pre-period industry composition and sectoral export growth may correlate with other determinants of rents and wages. These regressions do not identify the amenity elasticity used in the model.

3.6 The Evolution of Border Costs in Europe

The counterfactual represents enlargement as a reduction in border-related trade costs. I estimate separate border coefficients for trade within the EU15 (β^{WW}), within the new member states (β^{EE}), and between the two blocs (β^{EW}), using the gravity specification

In estimating border effects, I employ a traditional gravity specification with origin and destination fixed effects, given by

$$\log T_{nm} = \beta_0 + \beta_1 \log dist_{nm} + \beta_2 B_{nm} + \beta' \mathbf{X} + \delta_n + \delta_m + \epsilon_{nm} \tag{18}$$

[16] Of the 24 German hubs, 19 have the required data. A pooled hub/non-hub interaction test gives $p = 0.097$ for the difference in exposure coefficients.

Table 1: Export Exposure and Rent Growth Relative to Wage Growth

	All Cities Δ Rent – Δ Wage	Hubs Δ Rent – Δ Wage
Export Impact (2004–2014)	0.00651	0.59747**
	(0.11592)	(0.27177)
Dwellings/Capita (2001)	-20.44	-44.45
	(31.42)	(39.81)
Observations	64	19
R^2	0.008	0.307

Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Conventional standard errors. The outcome is rent growth minus wage-per-capita growth over 2005–2014, in percentage points. Export exposure is measured in thousands of constant 2004 euros per manufacturing worker over 2004–2014, using 2001 industry employment weights. Columns use 64 German cities and 19 selected hubs, respectively; both control for 2001 dwellings per capita.

where T_{nm} denotes total bilateral trade between origin n and destination m, $dist_{nm}$ denotes the distance in km between these locations, B_{nm} is the border dummy that takes on the value 1 if origin and destination are located in different countries, $\mathbf{X}$ denotes the matrix of bilateral control variables and δ_n, δ_m are origin and destination fixed effects respectively.

Post-enlargement estimates use the ERFT regional flows of Santamaría et al. (2021), covering NUTS2 regions in 24 European countries from 2011 to 2017. I average annual flows and estimate equation (18). The separate national-border term reflects the distinction between within-country and international trade emphasized by Santamaría et al. (2023). Before enlargement, comparable regional flows are unavailable, so I use country-level COMTRADE flows for 1996–2004, with bilateral controls and distances from CEPII. Because the pre- and post-enlargement regressions use different geographic units and periods, their coefficient difference may reflect those differences as well as the accession process. I use it as a maintained input to the quantitative scenario.

Table 2 reports the estimates. I hold the EU15–EU15 log border-cost term fixed at 1.496, its post-enlargement value, to isolate changes involving the new member states. The East–East trade coefficient changes from -2.970 to -2.255, while the East–West coefficient changes from -3.408 to -1.243. In the quantitative implementation, the absolute values of these coefficients enter directly as log border-cost terms in τ, without division by $\sigma - 1$. The East–West multiplier therefore falls from $\exp(3.408)$ to $\exp(1.243)$, an 88.5% reduction under this maintained mapping. This mapping differs from recovering structural iceberg

Table 2: Border Effects Before/After the Enlargement of EU

	West–West (1) Before & After Accession	East–East (2) Before	East–East (3) After	East–West (4) Before	East–West (5) After
Border Effect	-1.496***	-2.970***	-2.255***	-3.408***	-1.243***
	(0.212)	(0.668)	(0.313)	(0.291)	(0.269)
Controls	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes	Yes
Observations	25633	25	1311	38	6680
R^2	0.72	0.96	0.81	0.97	0.85
Data Source	ERFT	Comtrade	ERFT	Comtrade	ERFT
Geo. Unit	NUTS2	Country	NUTS2	Country	NUTS2

Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. The dependent variable is log bilateral trade. West denotes the EU15; East denotes the new member states. Regressions include log distance, bilateral controls and origin and destination fixed effects. ERFT standard errors are clustered by unordered country pair; country-level COMTRADE regressions use conventional standard errors. Column 1 uses the post-enlargement coefficient in both model equilibria. Pre- and post-period estimates differ in geographic unit and data source.

costs through the model's gravity coefficient relation; the 88.5% figure is a scenario input, not a separately estimated tariff-equivalent change.

4 Model Predictions and Reduced Form Empirical Evidence

The model predicts that lower trade barriers increase population concentration around hubs, with larger gains for hubs whose market access improves most. In the enlargement setting, proximity to the EU15–NMS border provides a measure of exposure to the newly integrated markets. I test whether population growth changes in the data follow these patterns without imposing the model's structure.

Two comparisons guide the analysis: hubs versus other locations, and locations near the accession border versus those farther away. The hub comparison tests the general concentration mechanism; the border comparison tests whether growth is stronger where market access gains should be greater.

Following Brülhart et al. (2012), I compare changes in population growth near and far

from the accession border using a difference-in-differences specification:

$$Pop.Growth_{it} = \beta_1(B_{X,i} \times A_t) + \alpha_i + \alpha_t + \epsilon_{it} \tag{19}$$

The outcome $Pop.Growth_{it}$ is annualized population growth in analysis unit i over 1991–2001 or 2001–2011, calculated as the decade percentage change divided by ten. The indicator $B_{X,i}$ equals one within X kilometers of the accession border, and A_t equals one for 2001–2011. Location and period fixed effects are α_i and α_t. Selected hubs containing several municipalities form one analysis unit; other municipalities remain separate. The sample has 36,326 units and 72,652 observations across eight countries, including 58 Slovenian historical units. Missing outcomes are not filled with model-accounting imputations. Border distance uses municipality representative points, the mean member distance for aggregated hubs, and a 2 km boundary-alignment tolerance. I cluster standard errors at NUTS3 level; aggregated hubs use the cluster of their largest 2001 member. The coefficient β_1 measures the additional change in population growth within the border zone.

A causal interpretation requires population growth near and far from the border to have evolved in parallel absent enlargement, with no other differential shocks correlated with border proximity. The two decade intervals provide only one pre-enlargement growth observation per location, so they cannot establish parallel pre-trends. The 2001–2011 interval also includes three years before accession. I therefore interpret these estimates as evidence on decadal growth patterns consistent with the model, rather than as a precise annual treatment effect of accession.

To test whether hub growth differs within and outside the border zone, I add hub interactions:

$$Pop.Growth_{it} = \beta_1(B_{X,i} \times A_t) + \beta_2(A_t \times Hub_i) + \beta_3(B_{X,i} \times A_t \times Hub_i) + \alpha_i + \alpha_t + \epsilon_{it} \tag{20}$$

The indicator Hub_i identifies a selected hub. The coefficient β_2 compares the change in growth of hubs and non-hub units outside the border zone. The triple interaction β_3 measures the additional hub/non-hub growth differential inside the zone. I estimate both specifications at 50, 100 and 150 kilometer cutoffs.

Table 3: The impact of EU Enlargement on Population Growth Trends

	Dep: Annualized Pop. Growth		
	(1)	(2)	(3)
	50Km	100Km	150Km
Border x Accession	0.280**	0.473**	0.344**
	(0.119)	(0.193)	(0.174)
Year FE	Yes	Yes	Yes
Location FE	Yes	Yes	Yes
Observations	72652	72652	72652
R^2	0.618	0.622	0.620

Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. NUTS3-clustered standard errors. Annualized growth is the decade population percentage change divided by ten; coefficients are percentage points. Accession denotes the 2001–2011 period, compared with 1991–2001. Each column uses the indicated distance cutoff for the accession border. The sample contains 36,326 units in eight countries, including Slovenia; multi-municipality hubs form single units.

Table 3 shows higher growth after enlargement near the accession border. The additional annualized growth is 0.280 percentage points within 50 kilometers (SE 0.119), 0.473 points within 100 kilometers (SE 0.193), and 0.344 points within 150 kilometers (SE 0.174). All three estimates are significant at the 5% level.

Table 4 shows that hubs outside the border zone gain 0.386–0.470 percentage points in annualized growth relative to non-hub units outside the zone. This hub differential is significant at the 1% level for all three cutoffs. Within 50 kilometers, the additional hub/non-hub differential is 0.498 points (SE 0.272; $p = 0.067$). The corresponding estimates within 100 and 150 kilometers are imprecise and not statistically significant. The general hub differential is stable across cutoffs; evidence for an additional near-border hub effect is weaker. The reported p-values do not adjust for testing the three cutoffs.

4.1 Robustness

Appendix A.1 examines whether the population pattern persists among cities alone and whether employment growth in Poland follows a similar geography. These comparisons use different samples and outcomes, so they assess the pattern's scope rather than repeat the

Table 4: Hub Population Growth and Proximity to the Accession Border

	Dep: Annualized Pop. Growth		
	(1)	(2)	(3)
	50Km	100Km	150Km
Border x Accession	0.279**	0.473**	0.344**
	(0.120)	(0.194)	(0.175)
Accession x Hub	0.386***	0.470***	0.426***
	(0.124)	(0.103)	(0.100)
Border x Accession x Hub	0.498*	0.246	0.325
	(0.272)	(0.280)	(0.273)
Year FE	Yes	Yes	Yes
Location FE	Yes	Yes	Yes
Observations	72652	72652	72652
R^2	0.618	0.623	0.620

Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. NUTS3-clustered standard errors. Annualized growth is the decade population percentage change divided by ten; coefficients are percentage points. Accession denotes the 2001–2011 period, compared with 1991–2001. Each column uses the indicated distance cutoff for the accession border. The sample contains 36,326 units in eight countries, including Slovenia; multi-municipality hubs form single units.

main specification.

Table A.1 uses exclusive 0–50, 50–100 and 100–150 kilometer bands and restricts the sample to DEGURBA-1 cities. Excluding Slovenia, as in the original appendix, leaves 308 units and 616 observations across seven countries. In column 1, the hub/non-hub differential outside the bands is −0.065 percentage points (SE 0.094), providing no evidence of a general hub advantage relative to other cities. The additional hub differentials are 0.865 points within 50 kilometers (SE 0.252) and 0.517 points at 50–100 kilometers (SE 0.153), both significant at the 1% level. The 100–150 kilometer estimate is imprecise. Column 2 replaces location fixed effects with hub-by-band terms to estimate pre-period level differences in growth. The first two are negative, followed by positive changes after enlargement. These pre-period differences are not a test of parallel growth trends.

As a final robustness check, I use regional data from Poland's national statistics office to test whether regions close to the German border experienced larger employment gains after 2004. For each NUTS3 region, the data report both total and urban employment, allowing me to estimate their growth responses separately. Table A.2 presents results for 66 NUTS3 regions and 1,320 annual growth observations from 1996 to 2015, with region and year fixed effects and NUTS3-clustered standard errors. The regressions include five exclusive distance rings through 250 kilometers, although only the first three are displayed. The accession interactions for the 0–50 and 50–100 kilometer bands are positive for both total employment (column 1) and urban employment (column 2); the 100–150 kilometer estimates are not statistically significant. These estimates show faster growth in both outcomes near the border. Because the regressions do not directly test a change in urban employment's share, I do not infer a change in urban concentration from this table alone.

5 Measuring the Effects of 2004 EU Enlargement

The border-cost counterfactual raises hub-core population by 0.465%, equivalent to 29.6% of its observed growth, and raises the average local real income by 0.22%. I first describe the counterfactual and assess model fit, then examine population concentration, income inequality and the amenity channel. Parameter sensitivity and a broader European scenario follow.

5.1 Isolating the Trade Channel of EU Enlargement

The counterfactual isolates enlargement's effect through integration in goods markets. I change border-cost terms involving the new member states while holding total population, exogenous amenities and the other structural parameters fixed. The exercise abstracts from demographic growth, changes in local fundamentals and other accession channels. I implement it as follows:

1. Solve the model for the pre-enlargement period, matching hub amenities and country specific amenities to 2001 census population data.
2. Compute the endogenous component of hub amenities related to trade.
3. Back out the exogenous component of hub amenities ($\bar{a}_h$).
4. Update border related barriers between NMS-EU15 (β^{EW}) and NMS-NMS (β^{EE}).
5. Solve the model for post-enlargement period, with reduced border related barriers, keeping the exogenous component of hub amenities ($\bar{a}_h$), country specific amenities ($\bar{a}_C$), total population in the entire sample and all of the structural parameters in the model fixed.

Fixed total population makes the counterfactual a spatial reallocation rather than a demographic forecast. Fixed country amenities preserve baseline country attractiveness, but country populations can change in response to relative returns.[17] The model allows workers to change country of residence and does not explicitly represent transitional labor-mobility restrictions. Germany and Austria retained restrictions until 2011, while Italy lifted them in 2006. This is a maintained limitation of the counterfactual, which changes goods-trade costs alone. Fixed exogenous hub amenities likewise exclude other local changes that could affect hub populations.

5.2 Model fit to municipality population data

I aggregate model population over the grid intersections within each municipality and compare the 2001 baseline with historical population levels. I also compare the border counter-

[17] Recalibrating country amenities to match post-enlargement country populations would instead prevent changes in country population totals.

factual with 2011 levels and 2001–2011 changes. The counterfactual keeps total population at its 2001 level; it is distinct from the auxiliary 2011 equilibrium used to identify α.

The 2001 population-level correlation is 0.952 across all municipalities and 0.488 outside selected hubs. For non-hub municipalities, the counterfactual population-level correlation with 2011 data is 0.484.[18] Correlations of population changes are lower: 0.265 for non-hub percentage changes, and 0.145 for absolute changes and 0.068 for log changes across all municipalities. The model captures population levels better than population changes. These comparisons assess fit rather than independently identify the trade channel, since observed changes also reflect forces excluded from the counterfactual.

5.3 Effects on Urban Primacy and Real Income

Hub-core population rises by 0.465% in the counterfactual, compared with 1.57% in the historical data. The modeled gain is 29.6% of observed hub growth. Hub cores' share of total population rises from 21.6047% to 21.7051%, a gain of 0.1004 percentage points. This is 130.7% of the observed share increase of 0.0768 points. The model therefore accounts for part of hub population growth while exceeding the observed rise in hub share; its fixed total population makes these distinct comparisons.

The population-weighted average local real-income gain is 0.22%, with a weighted standard deviation of 0.343 percentage points. Figure 4 maps changes in population-weighted municipality mean income. Locations using the same hub in both equilibria have the same percentage income gain; a small number switch hubs. Country mean income rises most in Slovenia (3.052%), followed by Slovakia (0.661%), Austria (0.654%) and the Czech Republic (0.629%). Germany gains 0.114% and Italy 0.059%. These country comparisons use each equilibrium's residential population weights, whereas the average local gain weights local percentage changes by counterfactual population. Real income excludes the amenity component of utility.

The model omits local land-use costs that could limit hub expansion. Evidence from port development shows that such costs can offset population gains from improved trade access (Ducruet et al., 2024). The size of the predicted hub-share response should be read in light of this omission.

[18] Of the 36,669 municipalities, 341 belong to the 73 hubs and 36,328 lie outside them. Fit comparisons include the model-accounting imputations described in Section 3.3. Only hub and country population totals are matched in the baseline calibration.

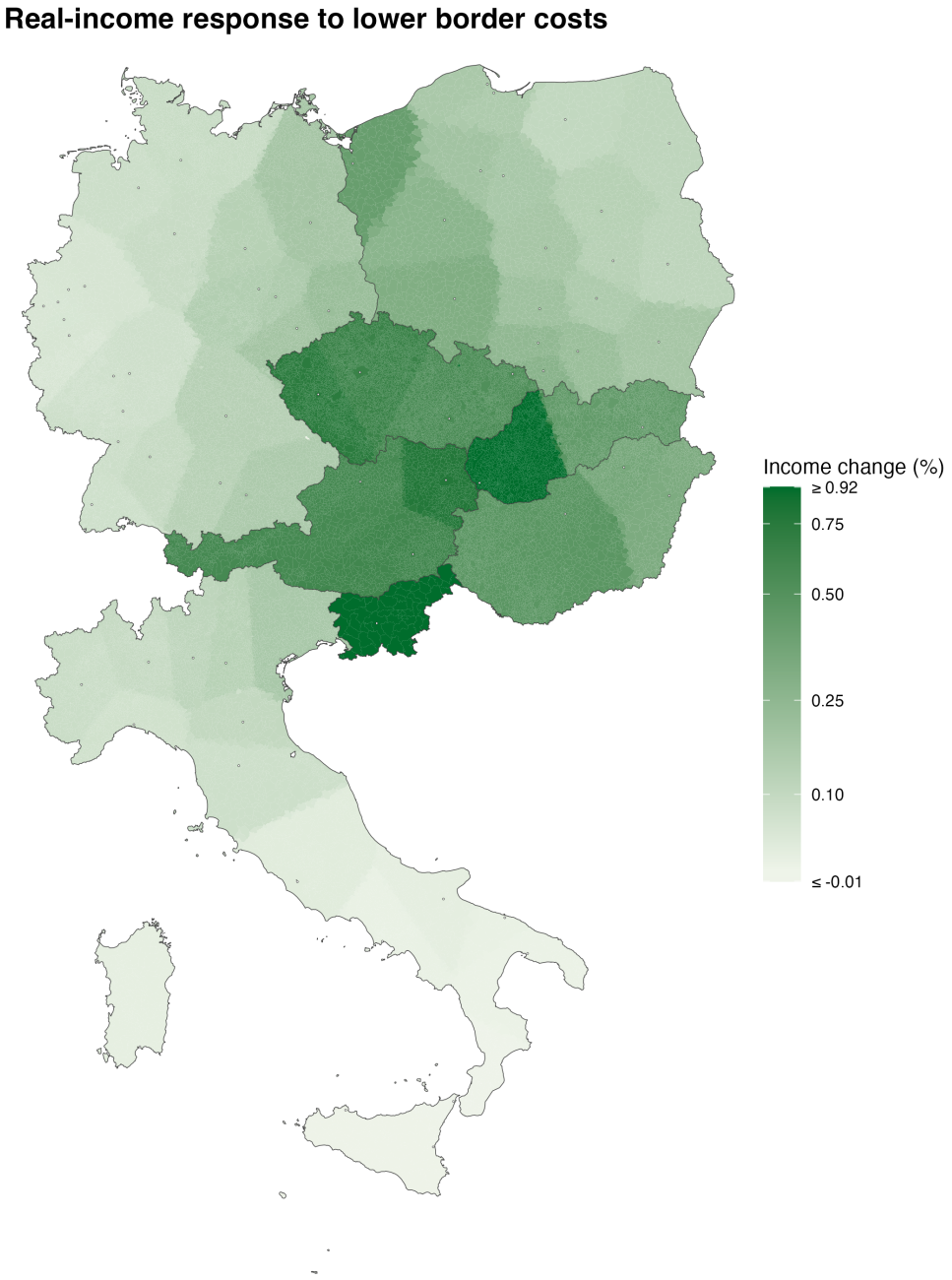


Figure 4: Map of real-income changes. Green shows gains in population-weighted municipality mean income; muted red shows losses. The nonlinear colour scale expands variation among gains and clips values beyond the labelled endpoints. Areas without a model result borrow a nearby colour for display only.

Urban growth can extend beyond city cores. Greater-city boundaries capture some cores spanning municipal borders, while functional urban areas (FUAs) include the surrounding commuting zone (Dijkstra and Poelman, 2014).[19] The latter geography is broader than the hub footprint in which the model places trade-related amenities.

To assess growth beyond hub cores, I overlay the associated 2014 Eurostat FUA polygons on the saved 1 km model locations.[20] The overlay is used only for reporting, with no role in calibration or the counterfactual solution. The modeled FUA population share rises by 0.088 percentage points, compared with 0.502 points in the data, accounting for about 17.5% of the observed gain. The FUA gain is smaller than the 0.1004-point gain in hub cores: areas inside these FUAs but outside the cores lose population on net. Core growth therefore drives the modeled FUA gain.

Figure 5 shows population concentrating around hubs, with larger gains in regions near the accession border. This geography is consistent with the reduced form growth patterns in Section 4. Fixed total population means that gains in these locations come with losses elsewhere. Germany and Italy illustrate how this reallocation interacts with existing regional income gaps.

5.4 EU enlargement and spatial inequality

Income gains vary across locations because enlargement changes their relative market access. Across the eight-country sample, the population-weighted standard deviation of local income falls by 0.431%. This aggregate decline coexists with increases in income dispersion within Germany and Italy.

In Germany, eastern hubs become more central to the integrated trade network. Mean income, weighted by population, rises by 0.164% in the East, including Berlin, and 0.092% in the West. The East–West gap narrows, but the weighted standard deviation across all German locations rises by 0.040%: convergence between broad regions need not reduce dispersion among individual locations.

Italy shows the opposite broad regional pattern. Mean income rises by 0.108% in the

[19] Eurostat defines the commuting zone using municipalities where at least 15% of employed residents work in the city; see `https://ec.europa.eu/eurostat/web/cities/methodology`.

[20] All 73 hubs lie in an FUA. There are 71 distinct polygons because two hub pairs share an FUA. I assign grid intersections by their representative points and allocate historical municipality population to intersections in proportion to land area.

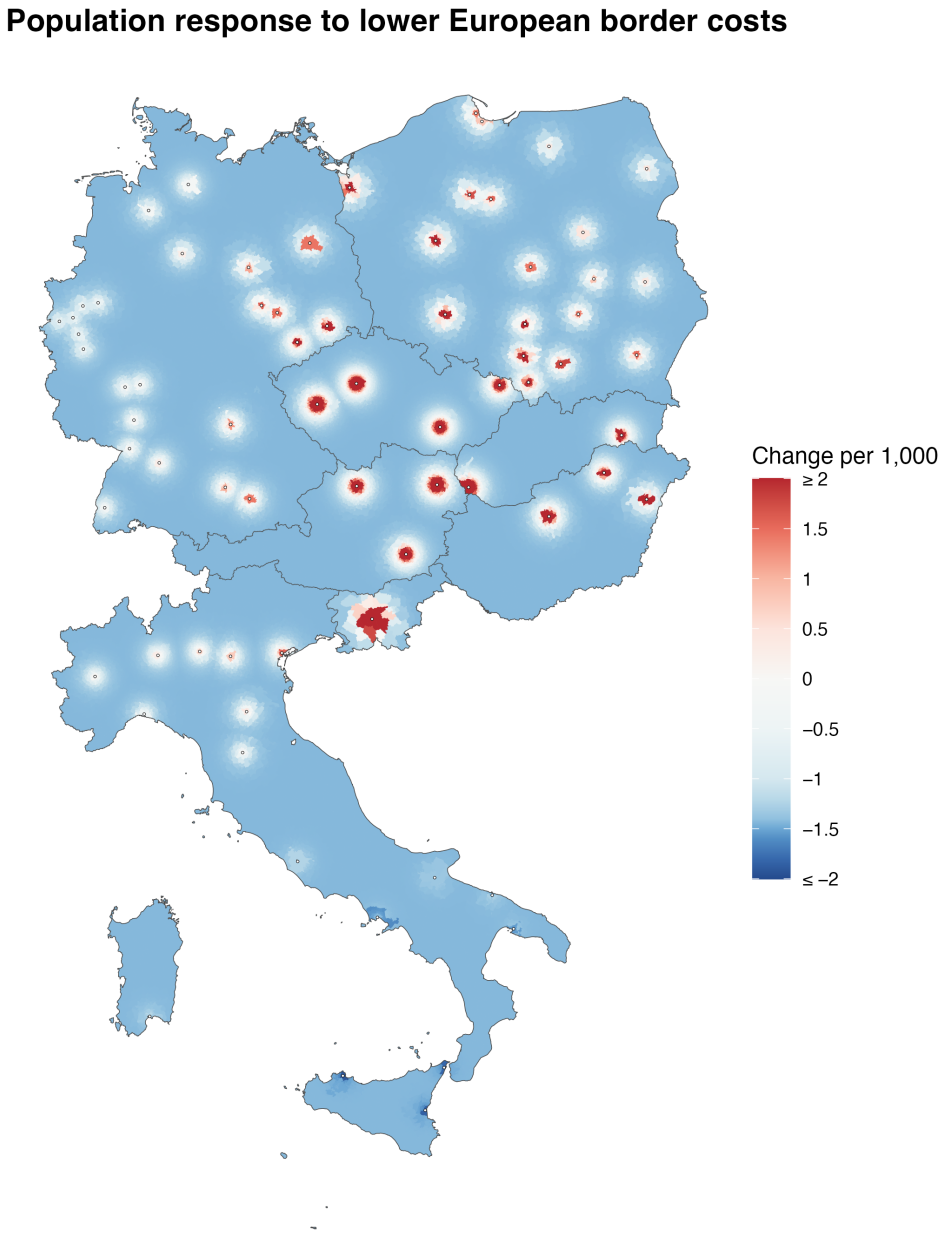


Figure 5: Map of population changes in the eight-country sample. Counterfactual change per 1,000 baseline residents by municipality. The fixed −2 to +2 colour range uses additional shades between its endpoints; values outside it are clipped for display. Areas without a model result borrow a nearby colour for display only.

North and 0.008% in the South, including the islands, widening the North–South gap.[21] The weighted standard deviation across Italian locations rises by 0.448%. The same integration shock can therefore narrow one regional divide while widening another.

5.5 Holding Hub Amenities Fixed

To measure the role of the amenity response, I repeat the border counterfactual while holding total hub amenities at their calibrated 2001 levels, $a_h = a_{h,2001}$. This retains baseline trade-generated amenities but prevents them from responding to the shock. Country amenities, total population and the other structural parameters remain fixed.

With hub amenities fixed, hub population rises by 0.186% and hub share by 0.0403 percentage points. Allowing amenities to respond adds 0.278 percentage points to hub population growth, accounting for 59.9% of the baseline gain. This comparison includes general equilibrium feedbacks; it is a counterfactual attribution rather than an additive causal decomposition. Average local real-income gains change little, from 0.215% with fixed amenities to 0.216% in the baseline (0.22% when rounded to two decimals). The amenity channel primarily amplifies population concentration, rather than real-income gains.

5.6 Robustness

I vary θ, ψ, ϕ and σ individually by 10%, reestimate the amenity elasticity α for each calibration, and repeat the border counterfactual. Table A.3 reports the resulting income, hub population and hub-share changes.

Changing taste heterogeneity θ, residence-to-hub shipping costs ψ or inter-hub transport costs ϕ leaves the population response close to the baseline. Across these six calibrations, hub growth ranges from 0.448% to 0.478%, and the hub-share gain from 0.097 to 0.103 percentage points. Greater taste heterogeneity weakens the incentive to relocate; higher shipping or transport costs also slightly reduce concentration gains. Real-income gains range from 0.201% to 0.234%.

Results are more sensitive to substitution across goods. A 10% increase in σ lowers the gains to 0.095% in real income, 0.334% in hub population and 0.072 percentage points in

[21] Northern Italy comprises the North-West and North-East NUTS1 areas; Southern Italy comprises the South and Islands. Central Italy gains 0.050%. Regional gains compare population-weighted mean income using each equilibrium's residential populations.

hub share. A 10% decrease raises them to 0.504%, 1.088% and 0.235 points, respectively. Greater substitutability reduces the gains from access to other locations' goods and hence the incentive to concentrate near hubs. The direction of the population response survives these perturbations, but its magnitude depends materially on σ.

5.7 A broader European sample

I also solve the original-rule model for 27 European countries and 172 selected hubs on a 5 km grid, using the eight-country estimate of $\alpha = 0.195$. The sample includes the United Kingdom, Switzerland and Norway. It applies the same east–west and east–east border-cost changes as the main exercise and groups Bulgaria, Romania and Croatia with the eastern countries, although they joined the EU after 2004. This is a common integration scenario, not an estimate of those later accessions.

Selected-hub population rises by 0.644% and its share of total population by 0.153 percentage points. Figure 6 shows gains around hubs, especially in Central and Eastern Europe and the Baltics, alongside losses in many surrounding municipalities. This is spatial reallocation under fixed total population.

The population-weighted average local real-income gain is 0.258%, while the weighted standard deviation of income across locations falls by 0.484%. Figure 7 shows larger gains in parts of the Baltics and Central Europe than in the western and southern periphery. Country mean income gains reach 4.229% in Slovenia, 3.961% in Latvia and 3.750% in Estonia, but are slightly negative in Spain, Portugal and the United Kingdom. The positive European average masks differences across countries. Because this scenario uses a broader geography and a coarser grid, it extends the spatial comparison rather than testing resolution convergence or estimating the 2004 accession effect alone.

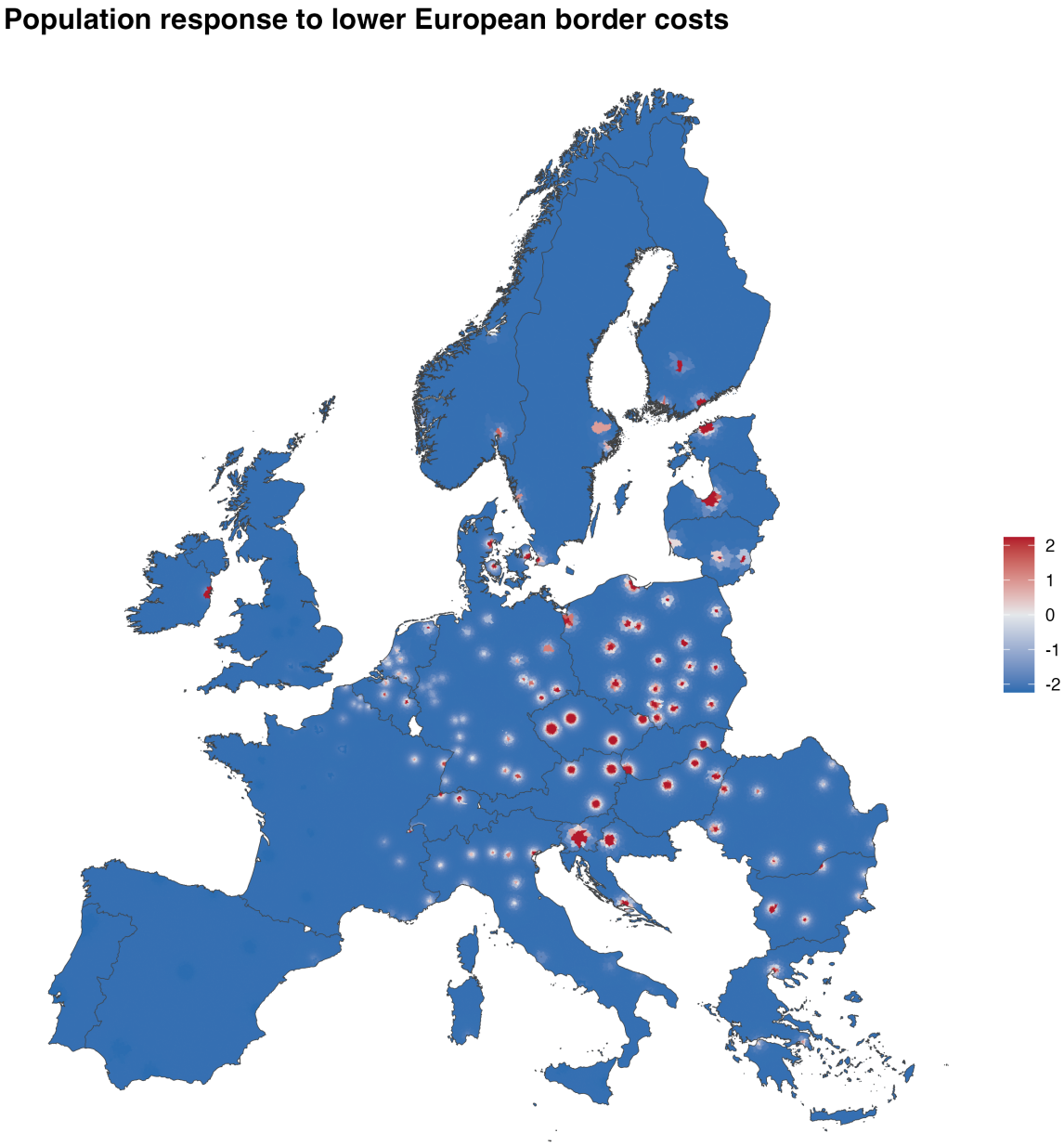


Figure 6: Population response in the broader European sample. Counterfactual change per 1,000 baseline residents by municipality; the symmetric colour scale is clipped at the 2nd and 98th percentiles. Areas without a population-backed model result borrow a nearby municipality colour for display only.

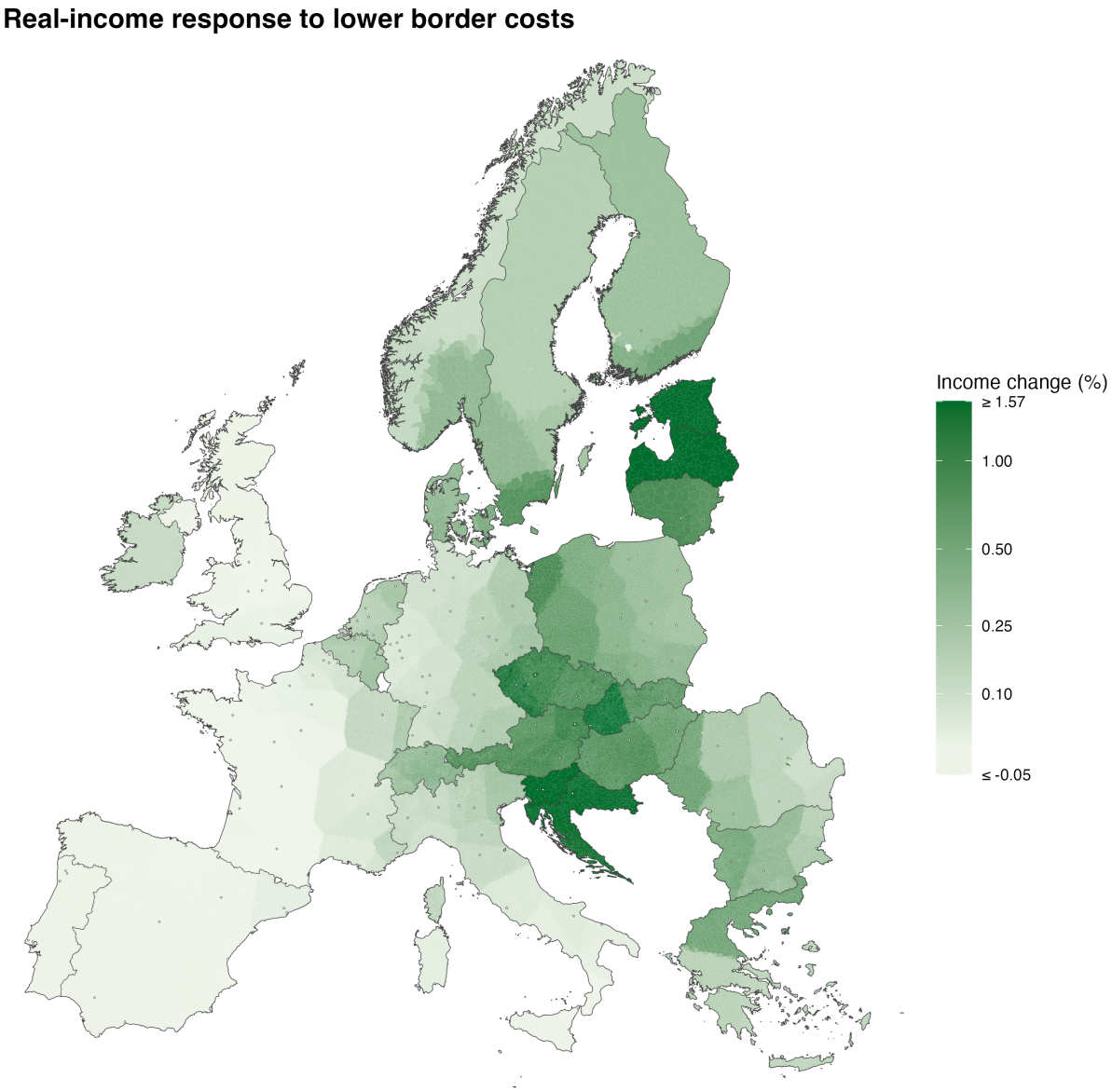


Figure 7: Real-income changes in the broader European sample. Green shows gains in population-weighted municipality mean income; muted red shows losses. The nonlinear colour scale expands variation among gains and clips values beyond the labelled endpoints. Areas without a model result borrow a nearby colour for display only.

6 Conclusion

European integration changes where people live as well as their real income. In the model, lower border costs raise the value of proximity to trading hubs, and trade-related amenities reinforce that incentive. The eight-country counterfactual generates a 0.465% increase in hub-core population, equivalent to 29.6% of observed hub growth. Its 0.1004 percentage-point increase in hub share exceeds the observed rise, while the broader FUA response accounts for 17.5% of the observed FUA share gain. Allowing hub amenities to respond to trade accounts for about 60% of the modeled hub-population increase.

Average local real income rises by 0.22%, but the spatial distribution of gains matters. Enlargement narrows Germany's East–West mean-income gap and widens Italy's North–South gap. Income dispersion falls across the eight-country sample while rising within both countries. Aggregate gains from integration therefore need not translate into convergence within every country; policies addressing regional disparities must account for these differences.

The quantitative results describe the goods-trade channel under fixed total population and the model's maintained border-cost mapping. The reduced form population and rent–wage patterns support the mechanism but do not identify every part of it. Two extensions would sharpen the analysis: direct measures of consumer amenities could test the trade-amenity response, and a model with housing supply and commuting could distinguish growth in city cores from expansion into their surrounding areas.

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A Data and Empirical Results

A.1 Additional Tables

Table A.1: Changes in Population Growth: Cities

	Dep: Annualized Pop. Growth	Dep: Annualized Pop. Growth
Acc x Hub	-0.0645	-0.0108
	(0.0938)	(0.0617)
Border(0–50) x Acc x Hub	0.8653***	0.8219***
	(0.2515)	(0.2449)
Border(50–100) x Acc x Hub	0.5167***	0.4733***
	(0.1527)	(0.1409)
Border(100–150) x Acc x Hub	0.4570	0.4136
	(0.4786)	(0.4760)
Border(0–50) x Hub		-0.4699**
		(0.1960)
Border(50–100) x Hub		-0.2232***
		(0.0784)
Border(100–150) x Hub		-0.4033
		(0.4617)
Year FE	Yes	Yes
Location FE	Yes	No
Observations	616	616
R^2	0.701	0.041

Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. NUTS3-clustered standard errors. The outcome is annualized population growth in percentage points over 1991–2001 and 2001–2011; Acc denotes the latter period. Border bands are mutually exclusive. The sample consists of 308 DEGURBA-1 city units in seven countries, excluding Slovenia. Column 1 includes location and period fixed effects; column 2 replaces location fixed effects with hub-by-band terms.

Table A.2: Poland (NUTS3): Changes in Employment

	Dep: Employment	Dep: Urban Employment
Border(0to50km) x Acc	0.01008**	0.01507***
	(0.00296)	(0.00376)
Border(50to100km) x Acc	0.00975**	0.01648***
	(0.00292)	(0.00386)
Border(100to150km) x Acc	-0.00264	0.00549
	(0.00453)	(0.00422)
Year FE	Yes	Yes
Location FE	Yes	Yes
Observations	1320	1320
R^2	0.765	0.636

Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. NUTS3-clustered standard errors. Outcomes are annual fractional growth in total and urban employment. Acc denotes years from 2005 onward. The sample contains 66 NUTS3 regions and 1,320 growth observations from 1996 to 2015. Distance bands are exclusive; the regressions also include the unreported 150–200 and 200–250 km rings. Region and year fixed effects are included.

Table A.3: Robustness: Main Results

	α	Income gain (%)	Hub population change (%)	Hub share change (pp)
Baseline Specification	0.195	0.216	0.465	0.100
10% higher heterogeneity parameter θ	0.201	0.216	0.448	0.097
10% lower heterogeneity parameter θ	0.187	0.216	0.478	0.103
10% higher shipping cost parameter ψ	0.195	0.216	0.458	0.099
10% lower shipping cost parameter ψ	0.194	0.216	0.466	0.101
10% higher transport cost parameter ϕ	0.198	0.201	0.459	0.099
10% lower transport cost parameter ϕ	0.191	0.234	0.467	0.101
10% higher elasticity of substitution σ	0.211	0.095	0.334	0.072
10% lower elasticity of substitution σ	0.165	0.504	1.088	0.235

Each parameter varies individually by 10%; the amenity elasticity α is reestimated in the eight-country sample. Income gains are local percentage changes weighted by counterfactual population. Hub population is population residing in selected hub cores; hub share is its share of total population, reported in percentage points. Entries are deterministic model counterfactuals, not estimates with sampling standard errors.

A.2 List of Hubs

City Code	City Name	Country
AT002C1	Graz	Austria
AT003C1	Linz	Austria
AT001C1	Wien	Austria
CZ002C1	Brno	Czechia
CZ003C1	Ostrava	Czechia
CZ004C1	Plzeň	Czechia
CZ001C1	Praha	Czechia
DE033C1	Augsburg	Germany
DE001C1	Berlin	Germany
DE034C1	Bonn	Germany
DE012C1	Bremen	Germany
DE505C1	Chemnitz	Germany
DE010C1	Dortmund	Germany
DE009C1	Dresden	Germany
DE011C1	Düsseldorf	Germany
DE006C1	Essen	Germany
DE005C1	Frankfurt am Main	Germany
DE027C1	Freiburg im Breisgau	Germany
DE018C1	Halle an der Saale	Germany
DE002C1	Hamburg	Germany
DE013C1	Hannover	Germany
DE035C1	Karlsruhe	Germany
DE004C1	Köln	Germany
DE008C1	Leipzig	Germany
DE019C1	Magdeburg	Germany
DE502C1	Mannheim	Germany
DE036C1	Mönchengladbach	Germany
DE003C1	München	Germany
DE014C1	Nürnberg	Germany
DE007C1	Stuttgart	Germany
DE020C1	Wiesbaden	Germany

City Code	City Name	Country
HU001C1	Budapest	Hungary
HU005C1	Debrecen	Hungary
HU002C1	Miskolc	Hungary
IT008C1	Bari	Italy
IT009C1	Bologna	Italy
IT029C1	Brescia	Italy
IT027C1	Cagliari	Italy
IT010C1	Catania	Italy
IT007C1	Firenze	Italy
IT031C1	Foggia	Italy
IT006C1	Genova	Italy
IT501C1	Messina	Italy
IT002K1	Milano	Italy
IT003K1	Napoli	Italy
IT005C1	Palermo	Italy
IT001C1	Roma	Italy
IT022C1	Taranto	Italy
IT004C1	Torino	Italy
IT011C1	Venezia	Italy
IT012C1	Verona	Italy
PL011C1	Białystok	Poland
PL506C1	Bielsko-Biała	Poland
PL008C1	Bydgoszcz	Poland
PL024C1	Częstochowa	Poland
PL006C1	Gdańsk	Poland
PL501C1	Gdynia	Poland
PL010K1	Katowice	Poland
PL012C1	Kielce	Poland
PL003C1	Kraków	Poland
PL009C1	Lublin	Poland
PL014C1	Olsztyn	Poland
PL005C1	Poznań	Poland

City Code	City Name	Country
PL025C1	Radom	Poland
PL015C1	Rzeszów	Poland
PL007C1	Szczecin	Poland
PL013C1	Toruń	Poland
PL001C1	Warszawa	Poland
PL004C1	Wrocław	Poland
PL002C1	Łódź	Poland
SI001C1	Ljubljana	Slovenia
SK001C1	Bratislava	Slovakia
SK002C1	Košice	Slovakia

A.3 Additional Figures



Figure A.1: Model Geography: Municipalities



Figure A.2: Model Geography: Market Areas in 2001. Each country has a distinct colour family, with a unique shade for every market; the shades identify markets and have no numeric order. Each displayed 1 km cell takes the market occupying the largest land area within it, agreeing with 99.880% of model intersection area. Reference areas outside the model borrow a nearby market colour for display only.

B Deriving Equilibrium Conditions

This appendix derives equations (7)–(11). Under CES demand, the price index at destination hub o is

$$P_o = \left[\sum_h p_h^{1-\sigma} L_h \tau_{ho}^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \tag{B1}$$

Each worker produces a distinct good and has one unit of labor, with one unit of labor required per unit of output. Because $\sigma > 1$, revenue increases with output; workers therefore produce their full unit endowment, $x^j = 1$ for every j. Workers trading through the same hub face the same demand and production conditions and set a common price, $p_h^j = p_h$. Substituting these conditions into equation (6) gives

$$p_h^\sigma = \sum_o \tau_{ho}^{1-\sigma} P_o^{\sigma-1} p_o L_o \tag{B2}$$

Given the Gumbel distribution of idiosyncratic amenities, the share of population living at residential location r is given by

$$\frac{L(r)}{\bar{L}} = \frac{[e^{a(r)+\max_h \zeta(\mu_h,r)^{-1} \frac{p_h}{P_h}}]^{\theta-1}}{\sum_s [e^{a(s)+\max_o \zeta(\mu_o,s)^{-1} \frac{p_o}{P_o}}]^{\theta-1}} \tag{B3}$$

The population share is the logit choice probability implied by the Gumbel tastes. All residents of location r choose the same trading hub $\mu(r)$. With hub real income $\omega_h = p_h/P_h$, equation (B3) becomes

$$\log L(r) = v + \theta^{-1}[a(r) + \zeta(\mu(r), r)^{-1} \omega_{\mu(r)}] \tag{B4}$$

Since by definition $\mu(r) = \text{argmax}_h \zeta(\mu_h, r)^{-1} \omega_{\mu_h}$, or the hub that provides the best economic incentives at location r, it must hold that

$$\zeta(\mu(r), r)^{-1} \omega_{\mu(r)} \geq \zeta(\mu_h, r)^{-1} \omega_{\mu_h} \quad \forall h. \tag{B5}$$

Using $P_h = \frac{p_h}{\omega_h}$, one can rewrite equation (B1) and (B2) as

$$p_h^{1-\sigma} \omega_h^{\sigma-1} = \sum_o \tau_{oh}^{1-\sigma} p_o^{1-\sigma} L_o \tag{B1'}$$

$$p_h^\sigma = \sum_o \tau_{ho}^{1-\sigma} p_o^\sigma \omega_o^{1-\sigma} L_o \tag{B2'}$$

Under symmetric inter-hub transport costs, I use the guess-and-verify approach of Allen and Arkolakis (2014). I normalize the common nominal price scale and conjecture

$$p_h = \omega_h^k$$

where k is a constant. I use the same nominal price and trade units in the numerical implementation, which applies the trade-related amenity T_h^α directly. No separate conversion of real income, amenities or θ is applied. Under this assumption, equations (B1') and (B2') become

$$\omega_h^{k(1-\sigma)+\sigma-1} = \sum_o \tau_{ho}^{1-\sigma} L_o \omega_o^{k(1-\sigma)}$$

$$\omega_h^{k\sigma} = \sum_o \tau_{ho}^{1-\sigma} L_o \omega_o^{k\sigma+1-\sigma}$$

The two equations coincide for $k = \frac{\sigma-1}{2\sigma-1}$

$$\omega_h^{\frac{\sigma(\sigma-1)}{2\sigma-1}} = \sum_o \tau_{ho}^{1-\sigma} L_o \omega_o^{-\frac{(\sigma-1)^2}{2\sigma-1}} \tag{B6}$$

Equation (B6), with coefficient one on its right-hand side, is the fixed point used in the numerical calculations. The reported calibration $\theta = 28.57$ is therefore used directly in the utility units of equation (B4).

Following Nagy (2022), I define the right-hand side of (B6) as hub market access MA_h. This gives

$$\omega_h = MA_h^{\frac{2\sigma-1}{\sigma(\sigma-1)}} \tag{B7}$$

Substituting this relation into (B4), (B5) and (B6) gives equations (7), (8) and (9), respectively.